

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

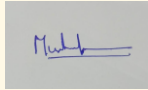
ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	SYED MUHAFIZ A	Reg. No.:	192472282
Section / Batch:		Date:	

Code of Conduct

I SYED MUHAFIZ A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. Morphological Processing for Search Engine

Input: *analyzing, analysis, analytical*

Morphological Decomposition

Word	Root	Affix(es)	Classification	Morphological Structure	Normalized Form
analyzing	analyze	-ing	Inflectional	analyze + ing	analyze
analysis	analyze	-sis	Derivational (verb → noun)	analyze + sis	analyze
analytical	analyze	-tic + -al	Derivational (noun → adjective)	analyze + tic + al	analyze

Summary

- **Inflectional:** analyzing
 - **Derivational:** analysis, analytical
 - **Common normalized root:** analyze
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2. Morphological Parsing for Sentiment Analysis

Input: *disagree, agreement, agreeable*

Morphological Breakdown

Word	Prefix	Root	Suffix	Classification	Meaning Change
disagree	dis-	agree	—	Derivational	Negates agreement
agreement	—	agree	-ment	Derivational	Verb → Noun
agreeable	—	agree	-able	Derivational	Verb → Adjective

Semantic Interpretation

- **dis-** adds negation.
- **-ment** forms a noun.
- **-able** forms an adjective indicating capability or suitability.

Normalized Root

Word	Morphological Structure	Normalized Root
disagree	dis + agree	agree
agreement	agree + ment	agree
agreeable	agree + able	agree

3. Morphology-Based Normalization

Input: *govern, government, governance*

Morphological Decomposition

Word	Root	Suffix	Classification	Morphological Structure	Normalized Form
govern	govern	—	Base Form	govern	govern
government	govern	-ment	Derivational	govern + ment	govern
governance	govern	-ance	Derivational	govern + ance	govern

Derivational Layers

- **-ment** → Forms a noun referring to an institution.
- **-ance** → Forms a noun referring to the process or system.

Normalized Representation

All words map to:

govern

4. Morphological Parsing for Document Classification

Input: *activate, activation, reactivation*

Morphological Breakdown

Word	Prefix	Root	Suffix	Classification	Morphological Structure	Normalized Root
activate	—	activate	—	Base Form	activate	activate
activation	—	activate	-ion	Derivational	activate + ion	activate
reactivation	re-	activate	-ion	Derivational	re + activate + ion	activate

Effect of Affixes

- **re-** indicates repetition ("again").
- **-ion** converts the verb into a noun.

Unified Base Representation

All forms normalize to:

activate

5. Morphology-Based Normalization for Search Optimization

Input: *create, creates, creating*

Morphological Analysis

Word	Root	Suffix	Classification	Morphological Structure	Normalized Output
create	create	—	Base Form	create	create
creates	create	-s	Inflectional (3rd person singular)	create + s	create
creating	create	-ing	Inflectional (present participle)	create + ing	create

Summary

- **Inflectional transformations:** creates, creating
- **Meaning remains unchanged; only grammatical form changes.**
- **Common normalized root:** create

Final Output

Word	Morphological Breakdown	Transformation	Normalized Form
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create	create	Base Form	create
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creates	create + s	Inflectional	create
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creating	create + ing	Inflectional	create
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Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____